

Comparing All Seven Parent Functions

TEACHER KEY | Algebra II Honors | Unit 1 | Day 8 | TEK A2.2.A

Objective: I can compare all seven parent functions across their key attributes, classify each as even, odd, or neither using an algebraic test, and state end behavior in formal notation.

Vocabulary

Even: $f(-x) = f(x)$ **Odd:** $f(-x) = -f(x)$ Most functions are **neither**.

Relative max / min — the largest or smallest output **on a given interval**, which may exist even when no overall one does.

Part 1 — The Master Table

Complete every cell from your Day 4-7 work. This is your reference for the year.

Function	Domain	Range	Intercept	Asymptotes
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$	$(0, 0)$	none
$ x $	all reals	$[0, \infty)$	$(0, 0)$	none
x^3	all reals	all reals	$(0, 0)$	none
$\sqrt[3]{x}$	all reals	all reals	$(0, 0)$	none
$1/x$	$x \neq 0$	$y \neq 0$	none	$x=0, y=0$
b^x	all reals	$(0, \infty)$	$(0, 1)$	$y = 0$
log base b	$(0, \infty)$	all reals	$(1, 0)$	$x = 0$

Part 2 — Even, Odd, or Neither

Substitute $-x$ into each rule. Do not decide from the picture alone.

Function	$f(-x)$ simplifies to	Even, odd, or neither
$ x $	$ x $	even
x^3	$-x^3$	odd
$\sqrt[3]{x}$	$-\sqrt[3]{x}$	odd
$1/x$	$-1/x$	odd
$\sqrt{x}$	undefined for $x < 0$	neither
2^x	2^x with $-x$, not equal to either	neither

$\sqrt{x}$ is "neither" for a different reason than 2^x . Explain the difference:

$\sqrt[3]{-x}$ is **not even defined** for positive x , while 2^x is defined but **matches neither test**.

Part 3 — Name That Function

- Odd, no intercepts, two asymptotes. $1/x$
- Even, minimum 0, domain all reals. $|x|$
- Domain $(0, \infty)$, range all reals, through $(1, 0)$. **log base b**

4. Odd, domain and range all reals, and steepens as x grows. x^3
5. Items 4 and its partner $\sqrt[3]{x}$ are both odd with the same domain and range. What single piece of information separates them?

One point, such as **(2, 8) versus (8, 2)**

Part 4 — Extremes on an Interval

- a. $f(x) = x^3$ on $[-2, 2]$: min **-8** max **8** b. $f(x) = |x|$ on $[-3, 1]$: min **0** max **3**
- c. $f(x) = 1/x$ on $[1, 4]$: min **1/4** max **1** d. $f(x) = 1/x$ on $[-1, 1]$: **neither - undefined at 0**

Item (d) is the interesting one. Why does a closed interval fail to guarantee a max and min here?

The function is **not defined on the whole interval** — there is a break at $x = 0$.

Part 5 — Going Further

1. Sort all seven by whether the domain is restricted. restricted: **$\sqrt{x}$, $1/x$, log base b**
2. Only two of the seven have a minimum. Name them and explain what they share.
- $\sqrt{x}$ and $|x|$ — **both have range $[0, \infty)$**
3. Must an even function pass through the origin? Give an example or counterexample.

No — **$f(x) = x^2 + 1$ is even but misses the origin**

Exit Ticket

Classify $f(x) = 1/x$ as even, odd, or neither, showing the algebra, and name one other parent function in the same class.

$f(-x) = 1/(-x) = -1/x = -f(x)$, so odd Another odd one: **x^3 or $\sqrt[3]{x}$**